A symmetry *breaking* event is when the conditions suddenly change such that the theory you would write down to describe how particles interact takes on a different, less symmetric, structure. After a symmetry breaking occurs, you can no longer swap around symbols in equations in the same way, and this change in symmetry expresses itself as altered behavior in the physical world.

Some of the symmetries we work with in physics are abstract and only obvious in the mathematics, but some are the usual stuff. Rotational symmetry is when something looks the same rotated by some angle (like a circle or a five-pointed star). Translational symmetry means something looks the same if you shift it to one side (e.g., a long picket fence moved over the distance of one picket, or a long straight line slid over by an inch). Breaking a symmetry involves doing something to the situation to make that symmetry no longer work. A wineglass has perfect rotational symmetry until a lipstick mark appears in one spot. A picket fence has translational symmetry until one of the slats is broken. Even a dinner party can include a symmetry breaking event, as is frequently observed by groups of physicists at conference banquets after the liquor comes out. As you wait patiently at the start of the meal, surrounded by a perplexing array of silverware, a small bread plate on either side of you, you are in a rotationally symmetric situation. As soon as one person reaches right or left to take up the bread plate, the symmetry is broken, and everyone else can follow their lead.[fn6]

No matter what kind of symmetry we're working with, as physicists, we'll see it in the equations describing the interactions. There are ways to encode rotational, reflection, and translational symmetries in equations, so you know the physics stays the same no matter how you rotate, flip, or move the system in question. Equations can also encode subtler kinds of symmetries best described using group theory and abstract algebra that are FASCINATING but sadly pretty far outside the scope of this book.

When electroweak symmetry breaking occurred, way back when the universe reached the ripe old age of 0.1 nanosecond, it was a kind of rearranging of the structure of physics on a fundamental level.[fn7] The rules particle interactions must follow are completely different in our post-electroweak-era universe. The previously vaporous Higgs field has become an ocean.

The water analogy isn't perfect. When you move through water, you're slowed down by drag, which means that you'll come to a halt if you stop